

Kunhee (Geoffrey) Kim

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SUMMARY

UC Berkeley Data Science graduate and data engineer who builds and scales production data pipelines, multi-terabyte data lakes and warehouses (PostgreSQL, Neo4j, Spark), and ML/NLP systems. Strong in robust ETL with strict data-integrity guarantees, SQL performance tuning, and Linux infrastructure. Bilingual in English and Korean.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Swift, Bash; PostgreSQL, Neo4j, Apache Spark, PostGIS, SQLite

Data Engineering: ETL/ELT pipeline design, data lake & warehouse architecture, schema design, idempotent ingestion, deduplication, performance tuning, web scraping

Python & ML/NLP: Pandas, Polars, FastAPI, SQLAlchemy, Pydantic; local LLM inference (Ollama), Graph Neural Networks, entity resolution, backtesting

Tools & Languages: Docker, Linux, cron/systemd, Git, Tableau; English (fluent), Korean (fluent)

EXPERIENCE

Global Key Advisors — Financial Data Analyst

Jan 2026 – Present · Bay Area, CA

- **Saved the firm from deploying a losing trading strategy** by exposing lookahead bias in a legacy sentiment model through rigorous out-of-sample backtesting — a perceived +2.39% edge collapsed to negative returns under realistic transaction costs.
- Replaced a fragile CSV-based SEC scraper with a resilient threaded PostgreSQL pipeline, **enabling the team to ingest and query the full universe of public filings** without file-size limits or data loss on restarts (28,000+ entities).
- Designed a modular PostgreSQL warehouse schema and tuned storage so analysts could **run complex cross-filing queries without performance bottlenecks** on tens of millions of records.
- Automated daily market-data ingestion from a remote Windows host to a Linux Spark cluster, **removing the firm's exposure to vendor IP bans and eliminating hours of manual alignment each week**.
- Engineered an unattended NLP pipeline that converts raw filing text into structured corporate-relationship graphs in Neo4j, **giving researchers queryable supplier/customer/acquirer edges** without manual review.

PROJECTS

Alpha Signals — Quantitative SEC Event Research Pipeline

github.com/kheekim02

Python, Polars, PostgreSQL, Neo4j, Ollama, Graph Neural Networks

- Designed an autonomous daily pipeline (signals → simulated trades → GNN predictions by 04:30 PDT) that lets researchers test whether SEC-filing corporate events predict stock returns — fully hands-off end to end.
- Modeled discrete corporate events as a Neo4j knowledge graph, enabling a Graph Neural Network to surface mispricings that flat text-sentiment models missed.

PathWise — Intelligent iOS Running App

Spring 2026 · github.com/kheekim02

Swift (SwiftUI, MapKit, ActivityKit), FastAPI, PostgreSQL + PostGIS, GraphHopper, Docker

- Built a routing engine combining GraphHopper with PostGIS to generate safe, scenic running loops with native turn-by-turn voice navigation, achieving **sub-2-second route generation**.
- Designed a database-agnostic fallback that kept routing available during outages, and **presented the app at Berkeley's startup pitch competition**.

LEADERSHIP & RESEARCH

Student Researcher, UC Berkeley — Prof. Jill Berrick (FamiliesNotFees)

2024 – 2025

- Used statistical modeling on national child-welfare data to show how charging low-income parents for foster care keeps children in the system longer, equipping state lawmakers with evidence to change the policy.
- Built Tableau dashboards and rebuilt the project website, **growing reach to 4,000+ views and 1,000+ unique visitors**.

Course Coordinator, ENGIN 270B — UC Berkeley Fung Institute

Summer 2025

- Built automated Excel tracking systems for a 100+ student graduate cohort, giving faculty actionable metrics to improve the curriculum.

EDUCATION

University of California, Berkeley — B.A. Data Science

May 2026

GPA 3.81 · Completed in 3 years · Berkeley Changemaker & Entrepreneurship & Technology (SCET) Certificates · 2-yr military service, Korea (2022–24)